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# Import libraries

import pandas as pd

import numpy as np


from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler


from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score


# Load dataset

df = pd.read_csv("h1.csv")


# a) Data Cleaning

# Remove duplicate rows

df.drop_duplicates(inplace=True)


# Numeric columns

numeric_cols = [

    'trestbps',

    'chol',

    'thalch',

    'oldpeak',

    'ca'

]


# Fill missing numeric values with mean

for col in numeric_cols:

    df[col] = df[col].fillna(df[col].mean())
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# Categorical columns
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```
categorical_cols = [
```

```
    'fbs',
```

```
    'restecg',
```

```
    'exang',
```

```
    'slope',
```

```
    'thal'
```

```
]
```

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# Fill missing categorical values with mode
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```
for col in categorical_cols:
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```
    df[col] = df[col].fillna(df[col].mode()[0])
```

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# b) Data Integration
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# Add new column
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```
df['hospital_id'] = np.random.randint(
```

```
    100,
```

```
    105,
```

```
    size=len(df)
```

```
)
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# c) Data Transformation
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# Convert text data into numbers
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```
encoder = LabelEncoder()
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```
transform_cols = ['sex','dataset','cp','restecg','slope','thal']
```

```
for col in transform_cols:
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```
    df[col] = encoder.fit_transform(
```

```
        df[col].astype(str)
```

```
    )
```

```

# Scale numeric data
scaler = StandardScaler()

df[numeric_cols] = scaler.fit_transform(
    df[numeric_cols]
)

# d) Error Correcting
# Function to remove outliers
def remove_outliers(data, column):

    Q1 = data[column].quantile(0.25)
    Q3 = data[column].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    return data[
        (data[column] >= lower)
        &
        (data[column] <= upper)
    ]

# Remove outliers from age column
df = remove_outliers(df, 'age')

# e) Data Model Building
# Create target column
df['target'] = (
    df['num'] > 0

```

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).astype(int)

# Input and output data
X = df.drop(
    ['target', 'num'],
    axis=1
)
y = df['target']

# Convert remaining text columns
object_cols = X.select_dtypes(include='object').columns
for col in object_cols:
    X[col] = encoder.fit_transform(
        X[col].astype(str)
    )

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2,random_state=42)

# Create model
model = LogisticRegression(max_iter=2000)

# Train model
model.fit(X_train, y_train)

# Predict output
y_pred = model.predict(X_test)

# Print accuracy
accuracy = accuracy_score(
    y_test,
    y_pred
)

print("Accuracy:", accuracy)

```